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Power tool with knurled bushing

Abstract

A power tool includes a housing, a drive assembly including an output shaft extending from the housing such that a tool element for performing work on a workpiece is attachable to the output shaft, and a bushing rotatably supporting the output shaft. The bushing includes an outer surface having a knurled texture.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Chinese Utility Model Application No. 202220295729.6 filed on Feb. 14, 2022, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

(2) The present invention relates to power tools, and more specifically to impact tools.

BACKGROUND OF THE INVENTION

(3) Impact tools, such as impact drivers and impact wrenches, are typically utilized to provide a striking rotational force, or intermittent applications of torque, to a tool element or workpiece (e.g., a fastener) to either tighten or loosen the fastener.

SUMMARY OF THE INVENTION

- (4) The present invention provides, in one aspect, a power tool including a housing, a drive assembly including an output shaft extending from the housing such that a tool element for performing work on a workpiece is attachable to the output shaft, and a bushing rotatably supporting the output shaft. The bushing includes an outer surface having a knurled texture.
- (5) In some embodiments, the bushing is insert-molded within the housing.
- (6) In some embodiments, the housing includes a motor housing portion and a gear case coupled to the motor housing portion, and the power tool further comprises a motor supported within the motor housing portion and a gear assembly supported within the gear case.
- (7) In some embodiments, the gear case includes a nose at a front end portion of the gear case, and wherein the output shaft extends through a bore in the nose.
- (8) In some embodiments, the bushing is positioned within the bore.
- (9) In some embodiments, the bore includes a corresponding knurled texture formed by insert molding the bushing within the gear case.
- (10) In some embodiments, the power tool includes a light assembly surrounding the nose.
- (11) In some embodiments, the light assembly is supported by a holder coupled to the nose, the nose includes a groove, and a retaining ring is received in the groove to retain the holder on the nose.
- (12) In some embodiments, the output shaft is an anvil, and the drive assembly further includes a camshaft and a hammer configured to impart consecutive rotational impacts upon the anvil in response to rotation of the camshaft.
- (13) In some embodiments, the anvil engages a rear end of the bushing such that the bushing both rotationally and axially supports the anvil.
- (14) In some embodiments, the bushing and the gear case are made of different materials.
- (15) In some embodiments, bushing is made of AISI 4140 or AISI 52100 steel, and the gear case is made of A380 aluminum.
- (16) In some embodiments, the knurled texture comprises a plurality of teeth, each tooth having the shape of a rectangular pyramid.
- (17) In some embodiments, each tooth has a height between 0.3 millimeters and 0.7 millimeters.
- (18) The present invention provides, in another aspect, a power tool including a housing with a motor housing portion and a gear case coupled to the motor housing portion, a motor supported within the motor housing portion, a gear assembly supported within the gear case and driven by the motor, and a drive assembly driven by the gear assembly. The drive assembly includes an anvil extending from the gear case, a camshaft driven by the gear assembly, and a hammer configured to impart consecutive rotational impacts upon the anvil in response to rotation of the camshaft. The power tool also includes a bushing insert-molded within the gear case. The bushing rotatably and axially supports the anvil.
- (19) In some embodiments, an outer surface of the bushing has a knurled texture.
- (20) In some embodiments, the knurled texture comprises a plurality of teeth, each tooth having a height between 0.3 millimeters and 0.7 millimeters.
- (21) In some embodiments, the bushing includes a cylindrical inner surface rotatably supporting the anvil and a groove formed in the inner surface, the groove retaining a lubricant.
- (22) In some embodiments, the housing includes a handle housing portion extending from the motor housing portion, and the handle housing portion includes a battery receptacle configured to receive a battery for powering the motor.
- (23) The present invention provides, in another aspect, a power tool including a housing with a motor housing portion and a gear case coupled to the motor housing portion, a motor supported within the motor housing portion, a gear assembly supported within the gear case and driven by the motor, and a drive assembly driven by the gear assembly, the drive assembly including an output shaft extending from the gear case. The power tool also includes a knurled bushing fixed within the gear case, the knurled bushing rotatably and axially supporting the output shaft.

(24) Other features and aspects of the invention will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an impact tool according to one embodiment.
- (2) FIG. 2 is a cross-sectional view of the impact tool of FIG. 1.
- (3) FIG. 3 is an enlarged cross-sectional view of a portion of the impact tool illustrated in FIG. 2.
- (4) FIG. 4 is a perspective view of a gear case of the impact tool of FIG. 1.
- (5) FIG. 5A is a perspective view of a bushing positioned on the gear case of the impact tool of FIG. 1.
- (6) FIG. 5B is a side view of the bushing of FIG. 5A.
- (7) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

- (8) FIG. 1 illustrates a power tool in the form of a rotary impact tool **10**. The impact tool **10** includes a housing **14** with a motor housing portion **18**, a front housing portion or gear case **22** (FIGS. 2-4) coupled to the motor housing portion **18** (e.g., by a plurality of fasteners), and a handle portion **26** extending from and disposed below the motor housing portion **18**. The handle portion **26** includes a grip **27** that can be grasped by a user operating the impact tool **10**.
- (9) Referring to FIGS. 1-2, in the illustrated embodiment, the motor housing portion **18** and the handle portion **26** are defined by two cooperating clamshell halves, each molded from a suitable plastic material. The impact tool **10** may also include a protective cover **29**, which surrounds and substantially encloses the motor housing portion **18** and the gear case **22**. In some embodiments, the cover **29** may be overmolded. In other embodiments, the cover **29** may be attached (and optionally, removably attached) to the motor housing portion **18** and the gear case **22** by a snap fit, a plurality of fasteners, by stretching the cover **29** over the housing **14**, or by any other suitable arrangement. The cover **29** is preferably made of a resilient material, such as rubber, to protect the tool **10** from damage if dropped, etc. In other embodiments, the cover **29** may be omitted.
- (10) With continued reference to FIGS. 1-2, the impact tool **10** has a battery pack **34** removably coupled to a battery receptacle **38** located at a bottom end of the handle portion **26**. The battery pack **34** includes a housing **39** supporting battery cells, which are electrically connected to provide the desired output (e.g., nominal voltage, current capacity, etc.) of the battery pack **34**. In other embodiments, the impact tool **10** may include a power cord for electrically connecting the impact tool **10** to a source of AC power. As a further alternative, the impact tool **10** may be configured to operate using a different power source (e.g., a pneumatic power source, etc.).
- (11) In the illustrated embodiment, the impact tool **10** includes a light assembly **41** located at a front end of the gear case **22** and surrounding a nose **43** of the gear case **22**. The light assembly **41** is supported by a holder **47** coupled to the nose **43** (FIGS. 1-2). In the illustrated embodiment, the holder **47** is at least partially disposed within the cover **29** of the housing **14**. The light assembly **41** is oriented so as to illuminate a workpiece during operation of the impact tool **10**. The illustrated light assembly **41** includes multiple, circumferentially-spaced light sources (e.g., LEDs), such that the light assembly **41** may advantageously illuminate the workpiece without shadows being created by a tool bit (not shown) attached to the impact tool **10**. The holder **47** may include transparent

covers/lenses associated with each of the light sources. The light assembly **41** preferably draws power from the battery pack **34** and may automatically illuminate during operation of the impact tool **10** and shut off after a predetermined time period following operation of the impact tool **10**.

(12) Referring to FIGS. 2-3, an electric motor **42** is supported within the motor housing portion **18** and receives power from the battery pack **34** when the battery pack **34** is coupled to the battery receptacle **38**. The motor **42** is preferably a brushless direct current (“BLDC”) motor having a rotor with a motor shaft **50**. A button or actuator **52**, extending laterally from the housing **14**, allows an operator to change the direction in which the motor **42** rotates the motor shaft **50**. The motor shaft **50** is rotatable about an axis **54** and is rotatably supported at its rear end portion by a bearing **102** (FIG. 3), which in the illustrated embodiment, is nested within a fan **58**. The fan **58** is coupled to the motor shaft **50** for co-rotation therewith (e.g., via a splined connection) behind the motor **42**. The impact tool **10** also includes a trigger **62** coupled to the handle portion **26** that is actuatable to energize the motor **42**.

(13) With reference to FIG. 3, the impact tool **10** further includes a gear assembly **66** coupled to the motor shaft **50** and a drive assembly or impact assembly **70** coupled to an output of the gear assembly **66**. The gear assembly **66** is enclosed within the gear case **22**. The illustrated gear assembly **66** includes a pinion **82** coupled to the motor shaft **50**, a plurality of planet gears **86** meshed with the pinion **82**, and a ring gear **90** meshed with the planet gears **86** and rotationally fixed within the gear case **22**. The planet gears **86** are mounted on a camshaft **94** of the drive assembly **70** such that the camshaft **94** acts as a planet carrier. Accordingly, rotation of the motor shaft **50** rotates the planet gears **86**, which then orbit along the inner circumference of the ring gear **90** and thereby rotate the camshaft **94** at a reduced speed and increased torque relative to the motor shaft **50**. The gear assembly **66** thus provides a gear reduction from the motor shaft **50** to the camshaft **94**. In other embodiments, the gear assembly **66** may be configured in any of a number of different ways to provide a gear reduction between the motor shaft **50** and an input of the drive assembly **70**.

(14) The drive assembly **70** of the impact tool **10** includes an output shaft in the form of an anvil **200** extending from the gear case **22**. The anvil **200** includes a bit holder **202** to which a tool element (e.g., a screwdriver bit; not shown) can be coupled for performing work on a workpiece (e.g., a fastener). The drive assembly **70** is configured to convert the continuous rotational force or torque provided by the motor **42** and gear assembly **66** to a striking rotational force or intermittent applications of torque to the anvil **200** when the reaction torque on the anvil **200** (e.g., due to engagement between the tool element and a fastener being worked upon) exceeds a certain threshold. In the illustrated embodiment of the impact tool **10**, the drive assembly **70** includes the camshaft **94**, a hammer **204** supported on and axially slidable relative to the camshaft **94**, and the anvil **200**.

(15) In the illustrated embodiment, a rear end portion of the camshaft **94** is rotatably supported by a bearing **98**, which in turn is supported by a rear portion of the gear case **22**. A front end portion of the camshaft **94** is rotatably supported by the anvil **200** (e.g., the front end portion of the camshaft **94** is sleeved within a bore formed in the rear end of the anvil **200**). The anvil **200**, in turn, is rotatably supported by a bushing **236**, which, as described in greater detail below, is fixed within the nose **43** of the gearcase **22**. In the illustrated embodiment, the anvil **200** also engages a rear end of the bushing **236**, such that the bushing **236** also supports the anvil **200** in the forward axial direction. In other embodiments, the camshaft **94** and/or the anvil **200** may be supported in other ways. For example, in some embodiments, the anvil **200** may include a boss extending into a bore in the front of the camshaft **94** to rotatably support the anvil **200** and/or the camshaft **94**.

(16) The drive assembly **70** further includes a spring **208** biasing the hammer **204** toward the front of the impact tool **10** (i.e., toward the left in FIG. 3). In other words, the spring **208** biases the hammer **204** in an axial direction toward the anvil **200**, along the axis **54**. A thrust bearing **212** and a thrust washer **216** are positioned between the spring **208** and the hammer **204**. The thrust bearing

212 and the thrust washer **216** allow for the spring **208** and the camshaft **94** to continue to rotate relative to the hammer **204** after each impact strike when lugs (not shown) on the hammer **204** engage with corresponding anvil lugs **220** and rotation of the hammer **204** momentarily stops. The camshaft **94** includes cam grooves **224** in which corresponding cam balls **228** are received. The cam balls **228** are in driving engagement with the hammer **204** and movement of the cam balls **228** within the cam grooves **224** allows for relative axial movement of the hammer **204** along the camshaft **94** when the hammer lugs and the anvil lugs **220** are engaged and the camshaft **94** continues to rotate.

(17) With reference to FIGS. 4-5B, the nose **43** of the gear case **22** includes a bore **240** (FIG. 4) that receives the bushing **236** (FIGS. 5A-B). In the illustrated embodiment, an outer surface of the nose **43** includes a groove **244** extending at least partially around a circumference of the nose **43**. The groove **244** is shaped and sized to receive a retaining ring **248**, such as a c-clip, o-ring, washer, or the like. The retaining ring **248** is configured to engage a corresponding groove on an interior surface of the holder **47** for the light assembly **41**, to thereby couple the holder **47** to the nose **43** of the gear case **22** (FIG. 3).

(18) Referring again to FIGS. 5A-B, the bushing **236** is generally cylindrical and includes an outer surface **236a** having a knurled texture (such that the bushing **236** may be referred to herein as a “knurled bushing”). In the illustrated embodiment, the bushing **236** is insert molded into the nose **43** of the gear case **22**, (that is, the gear case **22** is molded around the bushing **236**), such that a corresponding knurled texture is formed on an inner mating surface **242** of the gear case **22** (FIG. 4). In other embodiments, the gear case **22** may not include a projecting nose **43** but may instead include a generally flat front end surface, with a thicker portion defining the bore **240** for receiving the bushing **236**. As discussed below, the inventors have found that the knurled texture on the inner mating surface **242** of the gear case **22** and the outer surface **236** of the bushing **236** advantageously improves retention of the bushing **236** within the gear case **22**, while also reducing stresses on the gear case **22** and improving life cycle durability.

(19) The bushing **236** is preferably made of metal, and more preferably a high-toughness and wear-resistant alloy steel, such as AISI 4140 or AISI 52100 steel. The gear case **22** is preferably made of a metal suitable for molding (e.g., metal injection molding or die casting), such as A380 aluminum. The inventors have found that the combination of these two different materials increase the durability and lifespan of the bushing **236** and the gear case **22**. In addition, by forming the gear case **22** from aluminum, the gear case **22** is lighter in weight than if the gear case **22** were formed from steel, for example. In other embodiments, however, other materials and combinations of materials may be used.

(20) The knurled texture of the bushing **236** may comprise a plurality of rectangular pyramidal teeth projecting outwardly from the outer surface **236a**. In some embodiments, each tooth has a height between 0.3 millimeters and 0.7 millimeters. In some embodiments, the pitch between adjacent teeth is between 1.0 millimeters and 1.5 millimeters. In some embodiments, each tooth has a rounded tip or peak, with a radius between 0.05 and 0.2 millimeters. In other embodiments, the shape, height, and/or pitch of the teeth may vary.

(21) With reference to FIGS. 5A-5B, a cylindrical inner surface **236b** of the bushing **236** is smooth to provide a bearing surface that rotationally supports the anvil **200**. In the illustrated embodiment, the inner surface **236b** includes a groove **256** extending around an inner circumference of the bushing **236**. The groove **256** is positioned at a midpoint along the axial length of the bushing **236** and may be filled with a lubricant, such as grease, to reduce friction between the bushing **236** and the anvil **200**. In some embodiments, the bushing **236** may include multiple grooves **256**.

(22) Referring to FIGS. 1-3, in operation of the impact tool **10**, an operator depresses the trigger **62** to activate the motor **42**, which continuously drives the gear assembly **66** and the camshaft **94** via the motor shaft **50**. As the camshaft **94** rotates, the cam balls **228** drive the hammer **204** to co-rotate with the camshaft **94**, and the hammer lugs engage, respectively, driven surfaces of the anvil lugs

220 to provide an impact and to rotatably drive the anvil **200** and the tool element.

(23) After each impact, the hammer **204** moves or slides rearward along the camshaft **94**, away from the anvil **200**, so that the hammer lugs disengage the anvil lugs **220**. As the hammer **204** moves rearward, the cam balls **228** situated in the respective cam grooves **224** in the camshaft **94** move rearward in the cam grooves **224**. The spring **208** stores some of the rearward energy of the hammer **204** to provide a return mechanism for the hammer **204**. After the hammer lugs disengage the respective anvil lugs **220**, the hammer **204** continues to rotate and moves or slides forwardly, toward the anvil **200**, as the spring **208** releases its stored energy, until the drive surfaces of the hammer lugs re-engage the driven surfaces of the anvil lugs **220** to cause another impact.

(24) The bushing **236** rotatably supports the anvil **200** and also absorbs forward axial forces exerted on the anvil **200** (e.g., by the hammer **204**). The knurled texture on the bushing **236** provides a strong connection between the bushing **236** and the gear case **22** to better resist the forces on the bushing **236**. This may improve the stability of the anvil **200** and reduce anvil wobble. In addition, the knurled texture advantageously reduces stress concentrations between the gear case **22** and the bushing **236**, as compared to typical bushings, which include a cylindrical outer surface and which may be press-fit into the gear case. Because stress concentrations are reduced, the walls of the gear case **22**, particularly at the nose **43** of the gear case and the transition between the nose **43** and the remainder of the gear case, may be made thinner, resulting in both length and weight savings.

(25) Although the bushing **236** is shown incorporated into a rotary impact tool **10**, the bushing **236** may alternatively be used with other rotary power tools (e.g., drills, reciprocating saws, rotary hammers, pulse drivers, etc.) for supporting an output spindle or shaft. In such tools, the bushing **236** may substitute for a roller bearing, such as a needle bearing or a ball bearing, which may reduce the cost of the tool without reducing the lifespan of the tool.

(26) Various features of the invention are set forth in the following claims.

Claims

1. A power tool comprising: a housing; a drive assembly including an output shaft extending from the housing such that a tool element for performing work on a workpiece is attachable to the output shaft; and a bushing rotatably supporting the output shaft, wherein the bushing includes an outer surface having a knurled texture, the knurled texture including a plurality of teeth spaced apart along a length and a circumference of the outer surface such that the knurled texture provides a strong connection between the bushing and the housing to resist axial forces on the bushing, and wherein the bushing is insert-molded within the housing.
2. The power tool of claim 1, wherein the housing includes a motor housing portion and a gear case coupled to the motor housing portion, and wherein the power tool further comprises a motor supported within the motor housing portion and a gear assembly supported within the gear case.
3. The power tool of claim 2, wherein the gear case includes a nose at a front end portion of the gear case, and wherein the output shaft extends through a bore in the nose.
4. The power tool of claim 3, wherein the bushing is positioned within the bore.
5. The power tool of claim 4, wherein the bore includes a corresponding knurled texture formed by insert molding the bushing within the gear case.
6. The power tool of claim 3, further comprising a light assembly surrounding the nose.
7. The power tool of claim 6, wherein the light assembly is supported by a holder coupled to the nose, wherein the nose includes a groove, and wherein a retaining ring is received in the groove to retain the holder on the nose.
8. The power tool of claim 2, wherein the output shaft is an anvil, and wherein the drive assembly further includes a camshaft and a hammer configured to impart consecutive rotational impacts upon the anvil in response to rotation of the camshaft.
9. The power tool of claim 8, wherein the anvil engages a rear end of the bushing such that the

bushing both rotationally and axially supports the anvil.

10. The power tool of claim 2, wherein the bushing and the gear case are made of different materials.

11. The power tool of claim 10, wherein the bushing is made of AISI 4140 or AISI 52100 steel, and wherein the gear case is made of A380 aluminum.

12. The power tool of claim 1, wherein each tooth of the knurled texture has the shape of a rectangular pyramid.

13. The power tool of claim 12, wherein each tooth has a height between 0.3 millimeters and 0.7 millimeters.

14. A power tool comprising: a housing including a motor housing portion and a gear case coupled to the motor housing portion; a motor supported within the motor housing portion; a gear assembly supported within the gear case and driven by the motor; a drive assembly driven by the gear assembly, the drive assembly including an anvil extending from the gear case, a camshaft driven by the gear assembly, and a hammer configured to impart consecutive rotational impacts upon the anvil in response to rotation of the camshaft; and a bushing insert-molded within the gear case, the bushing rotatably and axially supporting the anvil, wherein the bushing includes an outer surface and a plurality of teeth spaced apart along a length and a circumference of the outer surface to resist axial forces on the bushing.

15. The power tool of claim 14, wherein the plurality of teeth defines a knurled texture.

16. The power tool of claim 15, wherein the knurled texture comprises a plurality of teeth, each tooth having a height between 0.3 millimeters and 0.7 millimeters.

17. The power tool of claim 15, wherein the housing includes a handle housing portion extending from the motor housing portion, and wherein the handle housing portion includes a battery receptacle configured to receive a battery for powering the motor.

18. The power tool of claim 14, wherein the bushing includes a cylindrical inner surface rotatably supporting the anvil and a groove formed in the inner surface, the groove retaining a lubricant.

19. A power tool comprising: a housing including a motor housing portion and a gear case coupled to the motor housing portion; a motor supported within the motor housing portion; a gear assembly supported within the gear case and driven by the motor; a drive assembly driven by the gear assembly, the drive assembly including an output shaft extending from the gear case; and a knurled bushing fixed within the gear case, the knurled bushing rotatably and axially supporting the output shaft, wherein the knurled bushing includes a plurality of teeth spaced apart along a length and a circumference of an outer surface of the bushing such that the plurality of teeth provides a strong connection between the bushing and the housing to resist axial forces on the bushing.
